

> = then click · ▼ = red triangle on that report's title bar · → = where the answer appears · reject H_0 when $p \leq \alpha$

SETTING UP

Open the data. File > Open > the .jmp file. A quantitative column must read **Continuous** — right-click the column header > Column Info > Modeling Type.

Categorical or quantitative? A number you could average is quantitative (weight, diameter, price, pounds). A label putting each unit in a group is categorical (accepted/rejected, smoker/non-smoker).

Define the parameter. Write " μ = the true mean [variable, with units] for all [population]." Name the mean, the units, and the population.

Hypotheses. $H_0: \mu = \mu_0$ always. "less / decreased" → $H_1: \mu < \mu_0$ · "greater / increased" → $\mu > \mu_0$ · "different / changed" → $\mu \neq \mu_0$. Never put $\bar{x}$ in a hypothesis.

Shape, $\bar{x}$, s, n. Analyze > Distribution > Y, Columns = variable > OK → histogram for shape; **Summary Statistics** for Mean, Std Dev, N, Std Err Mean.

TESTING ONE MEAN

t test (σ not given). Analyze > Distribution > Y = variable > OK > ▼variable > Test Mean > type μ_0 > leave the SD box empty > OK → **Test Mean** panel, **t Test** row.

z test (σ given). Same path, but also fill "Enter True Standard Deviation to do z-test rather than t test" → the panel now shows a **z Test** row.

Pick the right p-value. H_1 is $\neq$ → **Prob>|t|** · H_1 is $>$ → **Prob>t** · H_1 is $<$ → **Prob<t**. Ignore the Signed-Rank row.

No data, only summary stats. Distribution needs raw data, so compute by hand: $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$, $df = n - 1$. If they give s^2 , take the square root first.

Interpret the test statistic: "The sample mean is **t** standard errors above / below μ_0 , the value H_0 claims."

Interpret the p-value: "If μ really were μ_0 , there is a ___% chance of getting a sample mean at least this far from μ_0 by random sampling alone."

Conclusion, $p \leq \alpha$: "Since $p < \alpha$, we reject H_0 . There is sufficient evidence to conclude [H_1 in words]."

Conclusion, $p > \alpha$: "Since $p > \alpha$, we fail to reject H_0 . There is insufficient evidence to conclude [H_1 in words]." Never write "accept H_0 ."

CONDITIONS AND INTERVALS

Conditions for the t test. Independence: a random sample, less than 10% of the population, one measurement per unit. **Normality:** the problem states bell-shaped, or $n \geq 30$ so the CLT applies, or a symmetric histogram with no outliers.

Check normality in JMP. ▼variable > Continuous Fit > Normal > ▼Fitted Normal > Goodness of Fit → **Shapiro-Wilk Prob<W**; above α means normality is reasonable. Add ▼variable > Normal Quantile Plot and say the points follow the line inside the bands.

Confidence interval for μ . **Summary Statistics** already lists Upper / Lower 95% Mean. Other levels: ▼variable > Confidence Interval > 0.90 / 0.95 / 0.99. By hand $\bar{x} \pm t(\alpha/2, n-1) \cdot s/\sqrt{n}$.

Interpret it: "We are 95% confident the true mean [variable] for all [population] is between ___ and ___."

Probability about an average. $z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$, or Cols > New Column > Formula > Probability > Normal Distribution(a, μ , $\sigma/\sqrt{n}$) which gives $P(\bar{x} < a)$; subtract from 1 for "more than".

SAMPLING DISTRIBUTION & POWER

Describe the sampling distribution of $\bar{x}$ when H_0 is true. Give all three. **Shape:** bell-shaped / normal — because the population is normal, or by the CLT when $n \geq 30$. **Mean:** μ_0 . **Standard deviation:** $\sigma/\sqrt{n}$ (this is the standard error).

What values of $\bar{x}$ lead to rejection? critical value = $\mu_0 + (z)(SE)$, with $SE = \sigma/\sqrt{n}$ and z signed by the tail: left -1.645 , right $+1.645$ at $\alpha = .05$; two-tailed $\mu_0 \pm 1.96 \cdot SE$. Reject when $\bar{x}$ is past it.

β = P(Type II error). Re-standardize that same critical value using the **true** mean: $z = (\bar{x}_c - \mu_{true}) / (\sigma/\sqrt{n})$, then $\beta = P(\bar{x} \text{ lands in the fail-to-reject region})$.

Power = $1 - \beta$. "Probability of detecting this difference", "probability of rejecting H_0 when the true mean is ___" and "probability of correctly concluding" all mean power. "Probability of failing to reject" and "incorrectly failing to reject" mean β .

Worked example. $\mu_0 = 50$, $\sigma = 21$, $n = 36$, $\alpha = .05$, left-tailed. $SE = 21/6 = 3.5$; $\bar{x}_c = 50 - 1.645(3.5) = 44.24$. If $\mu_{true} = 43$: $z = (44.24 - 43)/3.5 = .354$, $\beta = .3613$, power = **.6387**.

Power from JMP. DOE > Design Diagnostics > Sample Size and Power > One Sample Mean, fill Alpha, Std Dev, Difference to Detect and Sample Size, leave **Power** blank > Continue. Leaving **Sample Size** blank instead answers "how big a sample do I need". It is two-sided by default, so trust the hand computation on one-tailed problems.

Sketch. Two normal curves sharing one axis: top centered at μ_0 with the tail past $\bar{x}_c$ shaded = α ; bottom centered at μ_{true} with the fail-to-reject side shaded = β and the other side = power. One vertical line at $\bar{x}_c$ through both.

Which error is possible? If you rejected H_0 , only a Type I error. If you failed to reject, only a Type II. Power rises when $n \uparrow$, $\alpha \uparrow$, $|\mu_{true} - \mu_0| \uparrow$, or $\sigma \downarrow$.

Type I in context: "We conclude [H_1 in words] when in reality [H_0 in words]." Probability α .

Type II in context: "We conclude there is not enough evidence for [H_1] when [H_1] is actually true." Probability β .

TWO PROPORTIONS — INDEPENDENT GROUPS

Type in a 2x2 count table. File > New > Data Table with three columns — Group (Nominal), Outcome (Nominal), Count (Continuous) — and one row per cell, e.g. (Help, Accepted, 222).

Analyze it. Analyze > Fit Y by X > Y = Outcome > X = Group > Freq = Count > OK → **Contingency Table** and **Tests**. Forgetting the **Freq** box makes JMP analyze only 4 observations.

Point estimate. ▼Contingency Table > Row %. Each $\hat{p}$ is the cell divided by **its own row total**, never the grand total; the estimate is $\hat{p}_1 - \hat{p}_2$. Example: $222/515 - 55/190 = .4311 - .2895 = .1416$.

Hypotheses. $H_0: p_1 - p_2 = 0$. If the rates are the same in both groups the difference is zero. "improves" → > 0 · "reduces" → < 0 · "different" → $\neq 0$.

Assumptions. Independently chosen random samples or random assignment; independence within each group (one response per person); and $n_1\hat{p}_1$, $n_1(1-\hat{p}_1)$, $n_2\hat{p}_2$, $n_2(1-\hat{p}_2)$ all ≥ 5 — which is just all four cell counts being at least 5.

Test and interval. ▼Contingency Analysis > Two Sample Test for Proportions → the difference, its standard error, **Prob>|z| / Prob>z / Prob<z**, and the confidence interval. **Pearson ChiSquare** in the Tests panel is the same two-sided test ($\chi^2 = z^2$).

By hand. pooled $\hat{p} = (x_1+x_2)/(n_1+n_2)$; $z = (\hat{p}_1-\hat{p}_2) / \sqrt{[\hat{p}(1-\hat{p})/(1/n_1+1/n_2)]}$. The interval is **not** pooled: $(\hat{p}_1-\hat{p}_2) \pm z\alpha/\sqrt{[\hat{p}_1\hat{q}_1/n_1 + \hat{p}_2\hat{q}_2/n_2]}$.

Interpret the p-value: "If the acceptance rate really were the same in both groups, there would be a ___% chance of seeing a difference in sample proportions at least this large."

TWO PROPORTIONS — PAIRED

Spot it. The same people measured twice, or matched pairs (before/after the video, younger/older siblings). Never use the two-proportion z test here — it throws away the pairing.

Label the table. **a** and **d** are the ties (no change); **b** and **c** are the discordant pairs that changed in each direction. Smoking: $a = 20$, $b = 5$, $c = 16$, $d = 9$, $n = 50$.

Ties, n' , T. ties = $a + d$, $n' = b + c$, and T is the count in the direction H_1 predicts. Smoking: ties = 29, $n' = 21$, $T = 16$. $H_0: p_b = p_c$, i.e. changes are equally likely both ways.

p-value, exact binomial. Under H_0 , $T \sim \text{Binomial}(n', 0.5)$. Cols > New Column > Formula > Discrete Probability > Binomial Distribution(.5, n' , k) gives $P(T \leq k)$. Smoking: $1 - \text{BinDist}(.5, 21, 15) = .0133 < .05$, so reject — the video was effective.

Or from JMP. Analyze > Fit Y by X > Y = After, X = Before, Freq = Count > OK > ▼ > Agreement Statistic → the **test of symmetry** is McNemar's test, but two-sided only.

Interval. $(b-c)/n \pm z\alpha/\sqrt{[(a+d)(b+c) + 4bc] / n^3}$. Smoking gives $(- .389, -.051)$, entirely below 0. Puzzle gives $(- .196, .077)$, which contains 0 — no significant difference.

CRITICAL VALUES & FORMULA FUNCTIONS

z critical values: one-tailed $\alpha = .10 \rightarrow 1.282$, $.05 \rightarrow 1.645$, $.025 \rightarrow 1.960$, $.01 \rightarrow 2.326$, $.005 \rightarrow 2.576$. Two-tailed 90% $\rightarrow 1.645$, 95% $\rightarrow 1.960$, 98% $\rightarrow 2.326$, 99% $\rightarrow 2.576$.

JMP formula functions: t Quantile($1-\alpha/2$, df), Normal Quantile($1-\alpha$), t Distribution(t, df) (left tail), $2*(1 - \text{t Distribution}(|t|, \text{df}))$ for a two-sided p-value.

JMP Exam Notes

CORRELATION & THE FITTED LINE

Which variable is which. "Predict A using B" means $y = A$ (response) and $x = B$ (explanatory). In JMP the response goes in **Y, Response** and the explanatory in **X, Factor**.

Scatterplot. Analyze > Fit Y by X > Y, Response = response > X, Factor = factor > OK. Comment on four things: direction, form, strength, and any unusual points. If the cloud bends, say a quadratic should be considered.

Add the line. **▼**Bivariate Fit > Fit Line → the equation is printed as text just above **Summary of Fit**.

Correlation r. Analyze > Multivariate Methods > Multivariate > both variables in Y, Columns > OK → **Correlations**, the off-diagonal value. Or $r = \pm\sqrt{\text{RSquare}}$ taking the sign of the slope. $|r| \geq .8$ strong, $.5-.8$ moderate, below $.5$ weak.

Equation of the least squares line. Analyze > Fit Model > Y = response > select x > Add > Run → **Parameter Estimates, Estimate** column: the Intercept row is b_0 , the x row is b_1 . Write it with real names, not y and x.

Predict. Substitute into $\hat{y} = b_0 + b_1x$. Watch units — if x is in thousands, 34,000 employees means $x = 34$. Don't extrapolate far outside the data.

Interpret the slope: "For each additional one [x-unit], the **predicted (average)** [y] increases / decreases by b_1 [y-units]." Say predicted or average, and never claim x causes y.

Interpret the intercept: "When $x = 0$, the predicted [y] is b_0 ." Add a caveat if $x = 0$ is outside the data or impossible.

CHECKING THE CONDITIONS

Save what you need. **▼**Response y > Save Columns > Residuals, then repeat for **Predicted Values** and **Studentized Residuals** → new columns at the far right of the data table.

Normality of the residuals. Analyze > Distribution > Y = Residual > OK > **▼**Residual > Normal Quantile Plot, then Continuous Fit > Normal > **▼**Fitted Normal > Goodness of Fit → **Shapiro-Wilk Prob<W**.

Residual plot. **▼**Response y > Row Diagnostics > Plot Residual by Predicted. A random band around zero means the conditions hold; an arch or U means a line is the wrong form; a funnel means the variance is not constant; studentized residuals past ± 3 are outliers.

Independence. Argue it from the design — a random sample of separate units is fine; be cautious when the data are consecutive years on the same place.

IS THE PREDICTOR USEFUL?

Test the slope. Fit Model > Run > Parameter Estimates, on the x row → **t Ratio** is the statistic, **Prob>|t|** the two-sided p-value, **Std Error** is $SE(b_1)$. $H_0: \beta_1 = 0$, $df = n - 2$. State H_0 , H_1 , t, p and a conclusion in context.

One-sided? Halve **Prob>|t|**, but only if the sign of b_1 agrees with H_1 ; if it disagrees the p-value is above $.5$ and you fail to reject.

Testing a slope against a number other than 0. JMP only tests $\beta_1 = 0$. Either check whether the confidence interval contains that number, or compute $t = (b_1 - c)/SE(b_1)$ with $df = n - 2$. "Do the data contradict a decrease of 2 per year?" → does the 99% CI contain -2 ?

Variation explained. Summary of Fit → RSquare. Say "___% of the variation in [y] is explained by the linear relationship with [x]."

THE THREE INTERVALS

CI for the slope. Right-click inside **Parameter Estimates** > Columns > Lower 95% + Upper 95% → read them on the **x** row, not the Intercept row. If it excludes 0, x is a useful predictor.

99% or 98% instead. **▼**Response y > Set Alpha Level > 0.01 (or **Other** → 0.02), then re-read those columns. If your build locks them at 95%, compute $b_1 \pm t(\alpha/2, n-2) \cdot \text{Std Error}$; get t from Formula > Probability > t Quantile($1-\alpha/2, n-2$).

CI for the mean response — "the average y for ALL cases with $x = \text{___}$ ". Add a data row with that x and **Y left blank**, re-run Fit Model > Run > **▼**Response y > Save Columns > Mean Confidence Interval → **Lower / Upper 95% Mean** on the row you added.

Prediction interval — "predict for ONE new case". Same setup, then Save Columns > Indiv Confidence Interval → **Lower / Upper 95% Indiv**. "Indiv" is always the wider one; both are centered on the same $\hat{y}$. Singular wording ("a truck", "a school", "a cafeteria") means Indiv.

Bands on the plot. **▼**Linear Fit > Confid Curves Fit or Confid Curves Indiv — graph only, no numbers.

The blank Y matters. A row added for a prediction must have an empty Y cell, or it joins the fit and moves the line.

Unusual points. Click the point to highlight its row; Rows > Exclude/Unexclude then re-Run to see the effect. "How many more votes than predicted" = actual — predicted = that row's residual.

CORRELATION · SIMPLE, MULTIPLE & POLYNOMIAL REGRESSION

MULTIPLE REGRESSION

Screen the predictors first. Analyze > Multivariate Methods > Multivariate > response + all predictors in Y, Columns > OK → **Correlations** plus a **Scatterplot Matrix**. The response row shows which predictor is most correlated with y; predictor-vs-predictor values above about $.8$ signal multicollinearity.

Fit it. Analyze > Fit Model > Y = response > highlight all predictors > Add > Run → **Parameter Estimates** gives $\hat{y} = b_0 + b_1x_1 + \dots + b_kx_k$.

Interpret a coefficient: "For each additional one [x_k -unit], the predicted [y] changes by b_k [y-units], **holding all other predictors constant.**" That last clause is where the points are.

Is the model useful? Analysis of Variance → F Ratio and Prob > F. H_0 : all $\beta = 0$; H_1 : at least one $\beta \neq 0$. Rejecting means at least one predictor is useful, not all of them.

Is one predictor useful, adjusting for the rest? Parameter Estimates, that predictor's row → **t Ratio, Prob>|t|**, $df = n - k - 1$. The **Effect Tests** table gives the same p-value as an F. A predictor can look strong alone and still be non-significant here.

CI for a coefficient. Right-click **Parameter Estimates** > Columns > Lower 95% + Upper 95%; set a different level first with **Set Alpha Level**. Keep "holding the others constant" in the interpretation.

Conditions. Save **Residuals** and **Studentized Residuals**, use **▼**Response y > Row Diagnostics > Plot Residual by Predicted, and run Distribution on the residuals for the histogram, NQ plot and Shapiro-Wilk. Multicollinearity numerically: right-click **Parameter Estimates** > Columns > VIF — above 5 is a warning, above 10 a problem.

CHOOSING A MODEL

Compare with adjusted R². Summary of Fit → RSquare Adj, highest wins. Plain R² always rises when a predictor is added, so it cannot compare models of different sizes; $R^2_{\text{adj}} = 1 - (1-R^2)(n-1)/(n-k-1)$ penalizes extras.

All possible models. Fit Model > add all > Personality = Stepwise > Run > **▼** All Possible Models → every subset with RSquare, RSquare Adj, Cp, AICc, BIC.

Backward selection. Fit Model > add all > Stepwise > Run, set Direction = **Backward**, Stopping Rule = **P-value Threshold**, Prob to Leave = **0.1**, click **Go → Current Estimates**, a check under **Entered** means kept. Then **Make Model → Run** for the full report. By hand: remove the predictor with the largest p-value if it exceeds 0.1, refit, repeat — one at a time.

Predict from the chosen model. Plug the values into that model's equation, or add a row with those x's and a blank Y and save **Predicted Values**, or use **▼**Response y > Factor Profiling > Profiler. Use the coefficients of the model you were told to use.

POLYNOMIAL (QUADRATIC) REGRESSION

Quick route. Analyze > Fit Y by X > Y, X > OK > **▼**Bivariate Fit > Fit Polynomial > 2, quadratic → a **Polynomial Fit Degree=2** block with the uncentered equation printed above it.

Full route (use this for intervals). Analyze > Fit Model > Y > select x > Macros > Polynomial to Degree with Degree = 2 > Run → **Parameter Estimates** gains an **x*x** row.

Centering. JMP centers polynomial terms by default, printing $(x - \bar{x}) \cdot (x - \bar{x})$ so the coefficients are not the raw b_0, b_1, b_2 . Uncheck **▼**Response y > Center Polynomials and Run again for the plain equation.

Is the quadratic useful? Test the squared term specifically: **Parameter Estimates**, the **x*x** row (or **Effect Tests**) → **Prob>|t|**. $H_0: \beta_2 = 0$ means a line is enough. The overall F can be significant just from the linear part, so don't use it here.

Change in y is no longer constant. change = $b_1(x_2 - x_1) + b_2(x_2^2 - x_1^2)$. From 0 to 1 that is $b_1 + b_2$; from 2 to 3 it is $b_1 + 5b_2$. The two differ — that is exactly the point of asking both.

Intervals from a quadratic. Add the row with x and a blank Y, fit with **Polynomial to Degree 2**, then **Save Columns → Mean Confidence Interval** for the average of all such cases, **Indiv Confidence Interval** for one specific new case.

REGRESSION FORMULAS

$S_{xx} = \sum(x - \bar{x})^2$ $S_{yy} = \sum(y - \bar{y})^2$ $S_{xy} = \sum(x - \bar{x})(y - \bar{y})$ · $b_1 = S_{xy}/S_{xx}$
 $b_0 = \bar{y} - b_1\bar{x}$ $r = S_{xy}/\sqrt{S_{xx} \cdot S_{yy}}$

df: $n - 1$ for one mean · $n - 2$ for a simple-regression slope · $n - k - 1$ for a multiple-regression coefficient · k and $n - k - 1$ for the overall F.